Scheduling Sports Equipment Production

1 Problem

Brendamore Sports produces footballs and soccer balls and must plan on how many to produce each month for the next six months. The tables in the datafile: Brendamore.xlsx contain demand forecasts, as well as the production costs and inventory holding costs, for the next six months. Brendamore must meet the monthly demand of each product through either a combination of inventory or production during the month. Assume that demand occurs at the end of the month, so that any production during a month can meet that month's demand.

During each month, there is enough production capacity to produce up to 32,000 total balls (football + soccer balls), and there is enough storage capacity to store up to 20,000 total balls (football + soccer balls) at the end of the month. Brendamore wants to meet these demands on time and it currently has 7,000 footballs and 5,000 soccer balls in inventory. In anticipation of demand beyond the six-month planning horizon, Brendamore would like to have 3,000 footballs and 3,000 soccer balls in inventory at the end of the sixth month.

2 Data

$M = \{Month\ 1, Month\ 2, Month\ 3, Month\ 4, Month\ 5, Month\ 6\}$

$B = \{Footballs, Soccer\ Balls\}$

$d_{ji} = Demand\ for\ product\ j\ in\ month\ i, for\ j \in B\ and\ i \in M$

$p_{ji} = Production\ cost\ of\ product\ j\ in\ month\ i, for\ j \in B\ and\ i \in M$

$h_{ji} = Holding\ cost\ of\ product\ j\ in\ month\ i, for\ j \in B\ and\ i \in M$

$initial\_inventory_j = Initial\ inventory\ for\ product\ j, for\ j \in B$

$final\_inventory_j = Required\ final\ inventory\ for\ product\ j\ at\ the\ end\ of\ month\ 6, for\ j \in B$

$production\_capacity = Monthly\ production\ capacity\ (in\ units)$

$inventory\_capacity = Monthly\ storage\ capacity\ (in\ units)$

3 Objective in Words

Decide how many units of footballs and soccer balls to produce each month to minimize total costs while meeting the demand for each product, subject to the following constraints:

- Production of footballs and soccer balls in each month ≤ 32,000 units.
- Ending inventory of footballs ≥ 3,000.
- Ending inventory of soccer balls ≥ 3,000.
- Non-negativity constraints apply for all production and inventory values.

4 Decision Variables

Let

$\boldsymbol{y_{ji}} = The\ number\ of\ units\ of\ product_j produced\ in\ each\ mount_i, for\ j \in B\ and\ i \in M$

$\boldsymbol{x_{ji}} = The\ inventory\ of\ product_j produced\ at\ the\ end\ of\ mount_i, for\ j \in B\ and\ i \in Mplus0.$

5 Algebraic Formulation

$$Minimize \sum_{j \in B} \sum_{i \in M} (p_{ji} . y_{ji} + h_{ji} . x_{ji})$$

(total_cost)

$subject\ to$

$\sum_{j \in B} y_{ji} \leq prod\ cpacity\ , i \in M$ (prod_capacity_constraint)

$\sum_{j \in B} x_{ji} \leq inventor\ capacity, i \in M$ (inventory_capacity_constraint)

$x_j, i - 1 + y_{ji} - d_{ji} = x_{ji}, j \in B\ and\ i \in M$ (inventory_balance_constraint)

$x_j, 0 = \text{initial inventory}_j, \text{j} \in \text{B}$ (initial_inventory_constraint)

$x_j, 6 = final\ inventory_j, j \in B$ (final_inventory_constraint)

$y_{ji} \geq 0\ , j \in B\ and\ i \in M$ (non-negativity Constraint)

$x_{ji} >= 0\ , j \in B\ and\ i \in Mplus0$

6 Implementation

An implementation and solution of the model using Python, AMPL is available here:

https://colab.research.google.com/drive/1SrO7HnNrF6ViCW0Xn9Sxr0uGSVvtTH4V?usp=sharing

7 Results

By the end of the six-month period, Brendamore Sports will have 3,000 footballs and 3,000 soccer balls in inventory, achieving an optimal total cost of $1,448,750.

Production Values (y):

Month	Footballs	Soccer Balls
1	16,000	5,000
2	17,000	15,000
3	20,000	10,000
4	5,000	5,000
5	2,500	5,000
6	8,000	10,500

Inventory Values (x):

Month	Footballs	Soccer Balls
1	7,000	5,000
2	8,000	0
3	0	0
4	0	0
5	0	0
6	3,000	3,000

8 ChatGPT Prompt:

https://chatgpt.com/share/66fe2a43-173c-800a-b1d2-2c98ee39d32e